

Problem #1

Due: Monday, October 13, 11:59 pm. Turn in to canvas, as described in the syllabus.

1. We start with a warm-up exercise in which you are asked to find the general solution of this system of linear, homogeneous, autonomous, first order differential equations by four different methods:

$$\begin{aligned}\dot{x} &= x + y + z \\ \dot{y} &= y + z \\ \dot{z} &= z\end{aligned}\tag{1}$$

- (a) Approach 1: Re-write as an equivalent set of three single-variable ordinary differential equations: one for $x(t)$ (third order), one for $y(t)$ (second order), and one for $z(t)$ (first order). For example, we already have that $\dot{z} = z$. You are asked to also find the second order equation for $y(t)$, which is of the form $\ddot{y} + a_1\dot{y} + a_0 = 0$, and the third order equation for $x(t)$. (Hint: for $\ddot{y}$, you will need to differentiate the $\dot{y}$ equation, and make the appropriate substitution for $\dot{z}$ in terms of $\dot{y}$ and y .) Use elementary methods to solve each of these differential equations to find $x(t)$, $y(t)$ and $z(t)$. Make sure to express any integration constants you have in terms of the initial condition $(x(0), y(0), z(0)) = (x_0, y_0, z_0)$.
- (b) Approach 2: The system of equations has a nice structure, where you could start with the $\dot{z}$ equation, which only depends on z and solve it for $z(t)$. You then plug that solution into the $\dot{y}$ equation which you can then solve using an integrating factor to obtain $y(t)$. Finally, plug your expression for $y(t) + z(t)$ into the $\dot{x}$ equation to solve for $x(t)$. Again, any integration constants should be in terms of the initial values of x, y, z .
- (c) Approach 3: Re-write (1) in matrix form as a differential equation for the vector valued function $\mathbf{V}(t) = (x(t), y(t), z(t))^T$, and find the general solution in terms of (generalized) eigenvectors of the matrix $\mathbf{A}$ that you obtain. Again make sure to express your general solution in terms of the initial condition (x_0, y_0, z_0) .
- (d) Approach 4: Write the solution as $e^{\mathbf{A}t}\mathbf{V}_0$, where $\mathbf{V}_0$ is the initial condition. You need to explicitly find the matrix $e^{\mathbf{A}t}$ so that you can evaluate this to check against your solutions obtained with the other approaches. All these methods must lead to the same result, afterall.
- (e) Short response: which approach did you like best? Which was the most tedious?
2. Consider the following linear, homogeneous differential equation

$$\frac{d\mathbf{X}}{dt} = \mathbf{M}(t)\mathbf{X}, \quad \mathbf{X} \in \mathbf{R}^2,\tag{2}$$

where

$$\mathbf{M}(t) = \begin{pmatrix} -1 + \gamma + \gamma \cos(2t) & 1 - \gamma \sin(2t) \\ -1 - \gamma \sin(2t) & -1 + \gamma - \gamma \cos(2t) \end{pmatrix}.$$

- (a) Compute the eigenvalues of $\mathbf{M}(t)$ and show that they do not depend on t . If these eigenvalues determined the stability properties of $\mathbf{X} = \mathbf{0}$, then how would the stability depend on the real parameter γ ?
- (b) Show that

$$\mathbf{X}_1(t) = \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix} e^{(2\gamma-1)t}, \quad \mathbf{X}_2(t) = \begin{pmatrix} \sin t \\ \cos t \end{pmatrix} e^{-t}$$

are linearly independent solutions to (2). Describe the possible asymptotic behavior of solutions as $t \rightarrow \infty$, as a function of the parameter γ . Are these solutions expressed in terms of the eigenvalues/eigenvectors of $\mathbf{M}$ for all γ , for any γ ?

- (c) Write the fundamental matrix solution, denoted $\Phi(t)$, to (2) that has as its initial condition $\Phi(0) = Id$, where Id is the 2×2 identity matrix. Briefly comment on it in light of general results of Floquet theory that apply to this problem. What is the monodromy matrix for this problem and what are the Floquet multipliers as a function of γ ? Briefly summarize your conclusions about the general solution of (2) based on what you learned in parts (a) and/or (b). For which values of γ do all solutions approach $\mathbf{X} = \mathbf{0}$ as $t \rightarrow \infty$?
 - (d) For this part, set $\gamma = \frac{3}{4}$, and plot the phase portrait associated with (2) for a sequence of values of t over one period T of the T -periodic matrix $\mathbf{M}$. (What is T ?) These phase portraits provide us with some snapshots for the instantaneous dynamics at different times. At each of these times t_j , also plot the “particle” locations $\mathbf{X}_1(t_j)$ and $\mathbf{X}_2(t_j)$ along with the direction that “particle” is heading. (You may have to do some zoom in or zoom out to be able to follow both solutions out to time T .) What do you think is the mechanism for a solution diverging to infinity even though we might have thought, based on the snapshots of the phase portrait, that solutions are spiraling into the origin? (This is open-ended and you may find yourself adding to it. I am wondering if I might generate other interesting examples. Perhaps I could generate another case by reversing time in this one. I wonder what are essential ingredients to get the instability. Could I make it more pronounced? Perhaps I should explore some other values of γ where the instability doesn't occur to check any conjecture I have. And so on. You are encouraged to generate a part (e) question to this problem set which you are not even required to answer!)
3. Suppose we want to minimize over x, y the function $f(x, y) = (x - a)^2 + (y - b)^2 + \epsilon n(x, y)$. A popular first-order minimization method is gradient descent: $(x_{k+1}, y_{k+1}) = (x_k, y_k) - \eta \nabla f(x_k, y_k)$.
- (a) When $\epsilon = 0$, what is the global minimum of $f(x, y)$? Is it unique?
 - (b) For what values of η (called the learning rate) is gradient descent a contraction map? For any initial condition (x_0, y_0) , will gradient descent converge?
 - (c) Still considering $\epsilon = 0$, write down Newton's method instead. How long (how many iterations) does it take to converge?
 - (d) Is Newton's method a contraction map?
 - (e) Now let $n(x, y) = \sin(2\pi x) \cos(2\pi y)$ and $\epsilon = 0.1$. What is the global minimum of $f(x, y)$?
 - (f) Is Newton's method still a contraction map?
 - (g) When is gradient descent a contraction map?

Always define the complete metric space (compact subsets of $\mathbb{R}^2$) on which a map may be a contraction. Based on your answers, ask yourself whether adding small perturbations to contraction maps can still be contraction maps.